
Hidden Pollution Regime Discovery Using a Discrete Hidden Markov Model

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Abstract

Air pollution measurements are influenced by weather conditions such as rainfall, but rain is not recorded in the UCI Air Quality dataset [6]. In this project we use a two-state hidden Markov model (HMM) to infer whether each hourly observation corresponds to a rainy or not rainy regime based solely on pollutant concentrations (CO , NO_2 , O_3). Rather than learning parameters, we construct a fixed HMM in which emission probabilities are derived from AQI categories and transition probabilities are set a priori. We generate synthetic ground-truth labels using Forward Filtering Backward Sampling (FFBS) and then apply Viterbi decoding to estimate the most likely sequence of latent regimes. Experiments with the dataset-derived emission probabilities show that the model collapses to a trivial solution and predicts only the not rainy state. When emission probabilities for low-AQI categories are increased, the HMM produces meaningful latent-state dynamics and achieves substantially higher precision, recall, and F1. These results highlight the sensitivity of HMM inference to emission design and illustrate both the strengths and limitations of using simple discrete HMMs to infer unobserved weather conditions from air quality observations.

1 Introduction

Air quality measurements fluctuate over time due to weather patterns, human activity, and atmospheric processes. While pollutant concentrations are recorded at one-hour intervals in the UCI Air Quality dataset [6], important meteorological conditions, such as whether it was raining, are not directly observed. Rainfall, however, has a measurable influence on pollutant dispersion and removal [4], leaving an indirect signature in the sensor data. This motivates the use of probabilistic time-series models that can recover unobserved weather conditions from observed pollutant levels.

In this project, we use a discrete Hidden Markov Model (HMM) to infer whether each hourly observation corresponds to a *rainy* or *not rainy* condition. The hidden states represent these two weather categories, and each state emits a discrete distribution over selected pollutant and meteorological variables from the dataset (CO , NO_2 , O_3). Because the dataset does not include rainfall measurements, we generate synthetic rain/no-rain labels using AQI-derived rain probabilities and known transition dynamics, and use these labels as a baseline for evaluating HMM performance. Our key questions are:

- How accurately can a simple discrete HMM recover rain/no-rain conditions from air quality observations?

- What emission patterns distinguish the inferred rainy and non-rainy states?
- How do transition dynamics reflect realistic temporal patterns in weather conditions?

Contributions. Although this is a course project, we structure our work in the style of a NeurIPS submission:

- We formulate rain/no-rain inference as a two-state discrete HMM problem over pollutant and meteorological features.
- We generate interpretable synthetic labels using AQI-category-based rainfall probabilities and state-transition assumptions.
- We evaluate the ability of the Viterbi-decoded HMM states to recover these labels and analyze the learned emission structure associated with rainy versus non-rainy conditions.

2 Related Work

Latent-state models such as Hidden Markov Models (HMMs) have been used to analyze air quality time series, where pollutant concentrations evolve according to unobserved atmospheric regimes. Prior work on $\text{PM}_{2.5}$ prediction [2] shows that HMMs with state-specific emission distributions can capture important temporal structure in pollution levels. Other studies model background exposure patterns using latent regimes [1], reinforcing the idea that hidden environmental conditions influence observed pollutant measurements. Extensions such as hidden semi-Markov models [7] address duration limitations of standard HMMs and motivate future improvements. In contrast to these forecasting-oriented or exposure-focused applications, our work evaluates whether a simple two-state discrete HMM can recover a weather-related hidden signal (rain versus no rain) using only pollutant-derived features.

3 Background

3.1 Air Quality Index (AQI) and Relevant Pollutants

The Air Quality Index (AQI) is a standardized metric that converts pollutant concentrations into a scale from 0 to 500, providing an interpretable summary of overall pollution severity. Although the AQI can incorporate several pollutants, three of its most influential components for this project are carbon monoxide (CO), nitrogen dioxide (NO_2), and ozone (O_3). These pollutants respond strongly to weather conditions such as rainfall, which can dilute or remove pollutants through atmospheric processes.

CO is produced by incomplete combustion and is often elevated during periods of high traffic or stagnant air. NO_2 originates from vehicular and industrial emissions and plays a key role in ozone formation. Ozone is a secondary pollutant formed by sunlight-driven reactions involving NO_x and volatile organic compounds. Because these pollutants vary in predictable ways under different atmospheric conditions, they offer indirect information about rainfall even when precipitation is not recorded.

In this project, we compute AQI categories from CO, NO_2 , and O_3 concentrations using official EPA breakpoint tables. These AQI categories are then mapped to estimated rainfall probabilities, which allows us to create synthetic rain labels for the Hidden Markov Model.

3.2 Hidden Markov Models

A Hidden Markov Model (HMM) represents a sequence of hidden states and corresponding observations. The model is defined by:

- a sequence of hidden states $z_{1:T}$,
- a sequence of observations $x_{1:T}$,
- an initial-state distribution π ,
- a transition matrix A , and

- emission parameters θ_k for each state.

The dynamics of the model are given by:

$$p(z_1) = \pi_{z_1}, p(z_t | z_{t-1}) = A_{z_{t-1}, z_t}, p(x_t | z_t) = p(x_t | \theta_{z_t}). \quad (1)$$

In our project, the hidden states represent rainy and not rainy conditions, and the observations consist of pollutant concentrations. We discretize observations using AQI categories, so each hidden state emits a categorical distribution over these discretized variables.

3.3 Learning and Inference

We do not learn HMM parameters; all transition and emission probabilities are fixed a priori. Using these fixed parameters, we generate synthetic hidden states with Forward Filtering Backward Sampling (FFBS) and then apply the Viterbi algorithm to obtain the most likely sequence. This allows us to study the model’s inferred regimes without performing parameter estimation.

4 Data

4.1 Dataset Description

Our primary dataset is the Air Quality dataset from the UCI Machine Learning Repository. It provides hourly measurements collected by chemical sensors in an Italian city and includes:

- pollutant concentrations such as CO, NO₂, O₃, NO_x, and NMHC,
- meteorological variables such as temperature, relative humidity, and absolute humidity, and
- timestamps at one-hour resolution.

Synthetic Rainfall Labels. Because the UCI Air Quality dataset does not include precipitation measurements, we generate synthetic rain and no-rain labels for evaluation. We first use NOAA precipitation records [3] to identify which calendar days experienced measurable rainfall. We then compute AQI categories for each hourly observation using CO, NO₂, and O₃ concentrations together with EPA breakpoint tables [5]. These AQI categories are mapped to estimated rainfall probabilities, which are combined with a fixed two-state transition model to sample a binary rain indicator at each time step.

Selected Features. We focus on CO, NO₂, and O₃ because these pollutants appear consistently in both the UCI and EPA datasets and are directly linked to AQI calculations. These variables form the observable sequence used by the HMM.

4.2 Preprocessing

We cleaned the UCI Air Quality dataset by removing invalid entries, standardizing pollutant units, and retaining only the variables used in our analysis. Numerical fields and timestamps were normalized for consistency. After preprocessing, we computed AQI categories from CO, NO₂, and O₃, mapped these categories to rainfall probabilities, and sampled a binary rain label for each hour using a fixed transition model. The final dataset therefore includes both the pollutant observations and the synthetic rain indicators used for HMM evaluation.

5 Modeling and Inference

5.1 Probabilistic Model

We consider a two-state hidden Markov model (HMM) for inferring rainy versus not rainy conditions from air quality measurements. At each hour t we introduce a latent regime variable

$$z_t \in \{0, 1\},$$

where $z_t = 0$ denotes a *not rainy* regime and $z_t = 1$ denotes a *rainy* regime. The observed variables are

$$x_t = (\text{CO}_t, \text{NO}_{2t}, \text{O}_{3t}) \in \mathbb{R}^3,$$

representing hourly concentrations of CO, NO₂, and O₃.

The HMM parameters are:

- Initial distribution π over z_1 .
- Transition matrix

$$A = \begin{bmatrix} 0.71428 & 0.285714 \\ 0.485981 & 0.514019 \end{bmatrix},$$

where $A_{ij} = P(z_t = j \mid z_{t-1} = i)$.

- Emission model defined in terms of an air quality index (AQI) category $c_t \in \{1, \dots, 6\}$ derived from x_t .

We construct a scalar AQI and corresponding category c_t from $(\text{CO}_t, \text{NO}_{2t}, \text{O}_{3t})$ using EPA break-point tables. Given a category c , we specify an empirical mapping

$$P(\text{rain} \mid c) = \begin{cases} 0.4040 & c = 1, \\ 0.3492 & c = 2, \\ 0 & c \geq 3, \end{cases}$$

and define the emission probabilities

$$P(x_t \mid z_t = 1) \propto p_c, \quad P(x_t \mid z_t = 0) \propto 1 - p_c,$$

where $p_c = P(\text{rain} \mid c_t)$. In practice we work with log-emission weights $\log e_t(1) = \log p_c$ and $\log e_t(0) = \log(1 - p_c)$. This encodes the assumption that lower AQI categories (better air quality) are more likely to coincide with rainy conditions.

5.2 Parameter Specification

We do not learn parameters from data. Instead, we fix them a priori:

- The transition matrix A is set to the empirical two-state transition rates given above.
- The initial distribution π is set to the stationary distribution of A , which satisfies $\pi = \pi A$.
- Emission weights are deterministic functions of the observed pollutants through the AQI construction and the mapping $P(\text{rain} \mid c)$.

These choices remove the need for maximum likelihood or EM training and isolate the effect of temporal inference under a fixed model.

5.3 Inference Algorithms

We use two standard HMM inference procedures.

Forward-Filtering Backward-Sampling (FFBS). Given the fixed transition matrix A and time-varying emission weights $\{e_t(z_t)\}$, FFBS first runs the forward algorithm to compute the filtered distributions $P(z_t \mid x_{1:t})$ for $t = 1, \dots, T$, then samples a complete latent sequence $z_{1:T}$ in reverse time using

$$P(z_t \mid z_{t+1}, x_{1:T}) \propto P(z_t \mid x_{1:t}) A_{z_t, z_{t+1}}.$$

We use FFBS to draw synthetic hidden state trajectories from the posterior of the fixed HMM.

Viterbi Decoding. To obtain a single most probable sequence of regimes, we apply the Viterbi algorithm. This dynamic program maximizes

$$\hat{z}_{1:T} = \arg \max_{z_{1:T}} P(z_{1:T} \mid x_{1:T})$$

by recursively storing the best predecessor for each state and time and then backtracking from the final time step. The result is a piecewise constant segmentation of the time series into rainy and not rainy regimes.

6 Experiments

6.1 Experimental Setup

We evaluate the HMM of Section 5 on the AirQualityUCI dataset. After cleaning and filtering, we obtain a chronologically ordered sequence of hourly CO, NO₂, and O₃ measurements spanning several thousand time steps.

For each hour t we:

1. Compute a scalar AQI and AQI category c_t from $(\text{CO}_t, \text{NO}_{2t}, \text{O}_{3t})$ using the EPA breakpoint tables.
2. Convert c_t into emission log-probabilities using the mapping $P(\text{rain} \mid c)$ defined in Section 5.
3. Apply FFBS under the fixed two-state HMM to sample a synthetic ground-truth sequence $z_{1:T}$ of rainy versus not rainy states.
4. Run Viterbi decoding on the same model to obtain the maximum a posteriori sequence $\hat{z}_{1:T}$.

Because the goal is to study temporal inference under a fixed generative model rather than predictive generalization, we treat the entire time series as a single sequence and do not perform train-validation splits.

6.2 Evaluation and Implementation Details

We evaluate sequence reconstruction by comparing the Viterbi path $\hat{z}_{1:T}$ to the FFBS sample $z_{1:T}$ using accuracy, precision, recall, and F1 score for the rainy state. All experiments use the fixed transition matrix

$$A = \begin{bmatrix} 0.71428 & 0.285714 \\ 0.485981 & 0.514019 \end{bmatrix}$$

and the emission mapping $P(\text{rain} \mid c)$ from Section 5. Log-emission probabilities are clipped to lie within $[10^{-6}, 1 - 10^{-6}]$ to avoid numerical underflow.

All inference routines (AQI computation, FFBS, Viterbi, and metric calculation) are implemented in Python using numpy and pandas, without external HMM libraries. A fixed random seed is used when sampling $z_{1:T}$ to ensure reproducibility.

7 Results and Discussion

7.1 Comparison of Model Configurations

We evaluated two configurations of the HMM. The first used the dataset-derived emission probabilities

$$P(\text{rain} \mid c) = \begin{cases} 0.4040 & c = 1, \\ 0.3492 & c = 2, \\ 0 & c \geq 3, \end{cases}$$

where c denotes the AQI category. Under this configuration, most observations fall into categories $c \geq 3$, which forces the emission probability for the rainy state to be zero for the majority of the dataset. As a result, both FFBS sampling and Viterbi decoding strongly favor the not rainy state.

In the second configuration, we manually increased the emission probabilities for low AQI categories and set

$$P(\text{rain} \mid 1) = 0.8, \quad P(\text{rain} \mid 2) = 0.6.$$

These values make the rainy state more plausible and allow both the sampler and the decoder to enter this regime more frequently. This configuration serves to illustrate how emission strength directly governs the expressiveness and stability of the inferred latent states.

7.2 Quantitative Results

Under the original emission mapping, the Viterbi decoder predicted the not rainy state for every time step. This produced an accuracy of 0.952 but zero precision, recall, and F1. Although the accuracy is high due to class imbalance, the model failed to recover any rainy intervals.

Under the hand-tuned configuration, increasing $P(\text{rain} \mid 1)$ and $P(\text{rain} \mid 2)$ produced substantial improvements:

$$\text{accuracy} = 0.929, \quad \text{precision} = 0.706, \quad \text{recall} = 0.649, \quad \text{F1} = 0.676.$$

FFBS generated 790 rainy labels under this configuration, and the Viterbi decoder produced 727 rainy predictions. These results show that the model can successfully recover the rainy regime when the emission model allows enough probability mass to support it.

7.3 Qualitative Insights

The original emission model leads to a collapse in which the latent sequence remains entirely in the not rainy state. This occurs because the emission probabilities forbid the rainy state for the majority of the dataset. As a result, neither FFBS nor Viterbi can produce meaningful latent-state structure.

When emissions are made more permissive, the inferred hidden states begin to align with expected patterns. Rainy segments appear primarily in regions where the AQI category falls into the lower range $c \in \{1, 2\}$, which is consistent with the model’s assumptions. However, the model still fails to identify rainy states at high AQI categories even when FFBS assigns them, which reflects a limitation of the categorical emission model.

Overall, the qualitative behavior shows that the HMM captures transitions between regimes only when the emission model provides sufficient flexibility. The latent structure is highly sensitive to the choice of emission probabilities.

7.4 Convergence, Stability, and Scalability

Viterbi decoding converged deterministically and was numerically stable in all experiments. Stability issues arise only when emission probabilities are exactly zero, which forces the latent state to collapse to a single regime. FFBS sampling is also stable when emission values avoid zero, but collapses when a state is forbidden by the emission model.

From a computational standpoint, the entire pipeline is efficient. Computing AQI values, running FFBS, and executing Viterbi decoding all scale linearly in the number of time steps T . The dataset used in this study contained approximately nine thousand time steps, and all computations completed almost instantly. The approach scales easily to much larger datasets.

Additional data would not improve model fitting since parameters are fixed, but more data would allow for richer posterior state sequences and more robust evaluation across seasonal or pollutant-related patterns.

8 Conclusion

We developed and evaluated a two-state HMM for modeling rainy versus not rainy regimes from air quality measurements. The results show that emission probabilities derived from AQI categories have a dominant effect on the latent sequence. When emission probabilities for the rainy state are zero for most AQI categories, the model collapses to a trivial solution. When emission probabilities are increased for lower AQI categories, the model produces meaningful latent-state transitions and achieves much better precision, recall, and F1.

These findings highlight the importance of selecting emission models that do not impose unrealistic or overly restrictive assumptions on the data. The model is flexible enough to recover hidden regimes when given reasonable emission parameters, but categorical emissions based solely on AQI categories may be too coarse to capture complex relationships between pollutants and rainfall.

Future work could include learning emission parameters through EM, replacing categorical emissions with Gaussian or mixture models, incorporating meteorological variables, or expanding the state space to capture additional environmental regimes.

9 Reflections and Contributions

9.1 Reflections and Advice for Future Students

Students attempting a similar project may benefit from the following observations. First, it is important to test multiple emission configurations early in the process. Emission probabilities that are exactly zero can force an HMM to collapse to a single latent state, which prevents meaningful inference. Inspecting intermediate outputs, such as the emission weights, FFBS samples, and Viterbi paths, can help identify these issues before running large experiments.

Second, students should explore several ways of mapping observations to emission probabilities. Categorical emission models are easy to implement but can be overly rigid. Continuous models or learned emission parameters often yield more expressive and realistic latent-state dynamics. It is also helpful to compare hand-tuned parameters with data-driven ones to understand how sensitive the HMM is to initialization choices.

Finally, students should begin with a simple two-state model before moving to more complex structures. Debugging and interpreting HMM behavior is much easier in low-dimensional settings, and insights gained from the two-state case often generalize to models with more states or richer observation spaces.

9.2 Team Contributions & Reflections

9.2.1 Contributions

- Benjamin implemented the core HMM inference pipeline, including FFBS sampling, Viterbi decoding, and the AQI-based emission model. He also led system integration, experimental analysis, and major portions of the final report.
- Miguel focused on data preprocessing, pollutant unit normalization, and constructing AQI categories for the emission model. He also helped evaluate experimental configurations and interpret the resulting latent-state sequences.
- Kanishk developed and tested the synthetic rainfall labeling pipeline, mapping AQI categories to rainfall probabilities and validating the transition model. He also contributed visualizations and analysis of model behavior across configurations.
- Derek supported implementation by reviewing code, troubleshooting numerical stability issues, and streamlining FFBS and Viterbi routines. He also assisted with metrics and formatting the results for the report.

9.2.2 Reflection

- Benjamin gained a deeper understanding of probabilistic sequence models and how modeling assumptions influence inference. He also learned the importance of careful debugging and validation when implementing HMM algorithms.
- Miguel found that working with real environmental data requires thorough cleaning and domain awareness, especially when transforming raw measurements into model inputs. He learned that small preprocessing choices can meaningfully affect downstream inference.
- Kanishk learned how synthetic-label generation shapes both evaluation outcomes and interpretability. He gained appreciation for the role of priors and parameter design in latent-state models when real labels are unavailable.
- Derek learned how numerical issues such as log underflow and zero-probability emissions can lead to inference failures. The project highlighted the importance of rigorous testing and practical considerations when implementing probabilistic models from scratch.

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